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RFI: Artificial Intelligence (AI) Action Plan

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Executive Summary

The Tomorrow Companies Inc. (Tomorrow.io) is a pioneer in the field of weather intelligence solutions, leveraging AI/ML, high-performance computing, and proprietary satellite data to revolutionize weather forecasting. We are uniquely positioned to advance the objectives of President Trump's Executive Order 14179, focused on American leadership in Artificial Intelligence.

Our AI-powered platform delivers actionable weather insights across all sectors of the economy, including defense, aviation, transportation, agriculture, wildfire prevention, emergency management, and logistics. We empower business and government agencies to proactively address weather-driven challenges, optimize resource allocation, and protect lives.

Key AI Policy Recommendations:

- **Empower Small Businesses As AI Innovation Leaders:** We advocate for policies that recognize and leverage the unique strengths of small businesses in driving AI innovation. Their agility, customer focus, and problem-solving capabilities are crucial for developing and deploying impactful AI solutions.
- **Establish Uniform Standards For Operational AI Models:** We urge the adoption of standardized metrics and procedures for AI model validation and calibration to ensure reliability, facilitate compliance, promote interoperability, mitigate risks, and support operational deployment.
- **Promote Data Licensing Agreements That Will Accelerate AI Growth:** We propose the implementation of clear, flexible, and accessible data licensing terms to fuel AI innovation. This includes policies that safeguard and promote companies at the forefront of developing proprietary data sets.

Tomorrow.io's expertise in AI and software engineering, combined with our proprietary satellite data and innovative solutions, positions us as a key partner in advancing the national AI strategy. We are committed to delivering weather intelligence that empowers informed decision-making, strengthens operational resilience, preserves resources, and safeguards lives.

The U.S. government should partner with the private sector to develop and deploy more advanced AI-enabled weather models and decision support tools to reduce risks to public safety and minimize property loss and damage to severe weather.

Section 1. Company Overview

Founded in 2016 in Boston, Tomorrow.io is a leading provider of innovative weather forecasting solutions, with extensive experience in Artificial Intelligence (AI)/Machine Learning (ML) weather prediction, software development, high-performance computing platforms, as well as numerical weather prediction. Our next-generation privately owned satellite constellation, combined with our deep engagement within the meteorological community, uniquely positions us to provide recommendations in support of the objectives of President Trump's signed Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence).

At Tomorrow.io, we are not just keeping pace with AI/ML advancements in weather modeling; we are actively driving them forward. Our intelligence-focused, adaptive solution threads advanced AI weather insights across all facets of business and government operations. Our proprietary satellite data, reinforcement-focused AI models, and Generative AI assistant collectively ensure that each decision—from defense missions to infrastructure planning—is grounded in an unparalleled understanding of evolving atmospheric conditions.

By continually learning from fresh observations and user feedback, Tomorrow.io's platform grows smarter with every forecast cycle, fostering unmatched operational resilience. Whether it's safeguarding soldiers, redirecting airline traffic, supporting farm yields, or rushing aid to disaster zones, the mission remains the same: empower government agencies to stay ahead of weather-driven challenges, preserve resources, and save lives.

Tomorrow.io has established itself as a leading global provider of Software-as-a-Service (SaaS) Weather Intelligence. Our Data-as-a-Service (DaaS) offer is designed to provide high-quality satellite data through subscription services, to our various customer segments. Tomorrow.io has acquired an impressive set of commercial and public sector customers including JetBlue, Uber, The National Football League, Boston Public Works, New York MTA, Washington DC MTA, NASA, NOAA, U.S. Air Force and U.S. Navy.

Section 2. AI Policy Recommendations

Small Businesses as a Driver of Innovation:

While large corporations dominate headlines with massive AI deployments, small businesses possess unique characteristics that position them as potent drivers of AI innovation. Their agility, close customer relationships, and focused problem-solving approach create a fertile ground for groundbreaking applications.

- *Nimble Experimentation & Rapid Prototyping:* Unlike large enterprises hampered by bureaucratic processes, small businesses can pivot quickly. They can rapidly test AI concepts, iterate based on immediate feedback, and deploy solutions without navigating layers of approval. This "fail fast, learn faster" environment fosters a culture of experimentation, crucial for exploring novel AI applications. Their limited resources often necessitate creative solutions, leading to innovative approaches that larger entities might overlook.
- *Deep Customer Intimacy & Targeted Solutions*: Small businesses often operate within niche markets, developing a profound understanding of their customers' specific needs and pain points. This close relationship enables them to identify precise problems that AI can solve, leading to highly targeted and effective solutions. They can tailor AI applications to address unique customer requirements, offering personalized experiences and optimizing processes in ways that mass-market solutions cannot.
- *Focused Problem-Solving & Resourcefulness:* With limited resources, small businesses are forced to prioritize and focus on solving specific, impactful problems. This constraint breeds resourcefulness and encourages them to leverage readily available AI tools and platforms. They are more likely to explore open-source AI libraries, cloud-based AI services, and no-code/low-code AI platforms, democratizing access to cutting-edge technology. This pragmatic approach leads to practical, real-world AI applications that deliver tangible results.
- *Embracing Niche Applications & Disruptive Potential*: Small businesses often operate in specialized sectors, making them uniquely positioned to develop niche AI applications that address industry-specific challenges. From AI-powered quality control in artisanal manufacturing to personalized learning platforms for specialized skills, they can leverage AI to disrupt traditional processes and create new market opportunities. These niche applications, while seemingly small, can aggregate into significant industry-wide transformations.
- *Collaborative Ecosystem & Community-Driven Innovation*: Small business owners are often deeply embedded in their local communities and industry networks. This fosters collaboration and knowledge sharing, creating a vibrant ecosystem for AI innovation. They are more likely to participate in local hackathons, workshops, and meetups, exchanging ideas and building partnerships. This collaborative spirit drives the development of innovative AI solutions that benefit the entire community.

While large corporations possess the resources for large-scale AI deployments, small businesses are the breeding ground for AI innovation. Their agility, customer intimacy, and focused problem-solving approach enable them to rapidly develop and deploy practical, impactful AI solutions. By embracing readily available AI tools and fostering collaborative ecosystems, small businesses are poised to lead the next wave of AI innovation, transforming industries and shaping the future of business. This administration should promote policies that support small businesses and position small businesses as leaders of AI innovation.

Uniform Standards for Operational AI Models

The rapid proliferation of artificial intelligence (AI) across diverse sectors, from healthcare and finance to transportation and governance, necessitates the urgent establishment of comprehensive standards for operational AI models. While the potential benefits of AI are undeniable, the absence of robust standards leaves us vulnerable to unpredictable performance, ethical breaches, and a lack of public trust. Establishing standardized frameworks for calibration, validation, and Technology Readiness Levels (TRLs), among other aspects, is crucial for responsible and effective AI deployment.

Calibration and Validation: Ensuring Reliability and Accuracy

AI models, particularly those based on machine learning, are inherently data-driven. Their performance is heavily reliant on the quality and representativeness of the training data. Without proper calibration, models can exhibit biases, inaccuracies, and a lack of generalizability. Standardized calibration procedures would ensure that models are fine-tuned to accurately reflect real-world conditions, mitigating the risk of systematic errors.

Similarly, rigorous validation is essential for assessing the model's performance and robustness. Standardized validation protocols would define clear metrics for evaluating accuracy, precision, recall, and other relevant performance indicators. This would not only provide a quantifiable measure of the model's effectiveness but also allow for comparisons across different AI systems. Standardized validation would also mandate stress testing to understand the performance boundaries of the model, and to identify potential failure modes. This is especially critical in safety-critical applications.

Technology Readiness Levels (TRLs): Guiding Development and Deployment

The concept of Technology Readiness Levels (TRLs), originally developed by NASA, provides a valuable framework for assessing the maturity of a technology. Applying TRLs to AI models would establish a clear roadmap for development, from initial research to full-scale deployment. Standardized TRL definitions for AI would provide a common language for researchers, developers, and regulators to communicate about the progress and readiness of AI systems.

For instance, a model in TRL 3 might demonstrate basic functionality in a controlled environment, while a model in TRL 9 would be fully integrated and operational in its intended environment. This framework would facilitate a more structured and transparent approach to AI development, ensuring that models are deployed only when they have reached an appropriate level of maturity and reliability. It would also allow for better risk management, by identifying potential challenges at each stage of development.

The need for standards extends beyond technical aspects to encompass ethical considerations and accountability. Standardized frameworks should address issues such as data privacy, algorithmic bias, transparency, and explainability. These standards must also guide the creation of clear lines of responsibility for the development, deployment, and maintenance of AI models. Furthermore, standardized audit trails and documentation requirements would enhance accountability and facilitate independent verification of AI system performance. This would build public trust and ensure that AI systems are used responsibly and ethically.

Establishing effective standards requires a collaborative effort involving researchers, developers, industry stakeholders, policymakers, and the public. International cooperation is also crucial to ensure consistency and interoperability across different jurisdictions. By establishing robust standards for operational AI models, we can unlock the full potential of this transformative technology while mitigating its risks. This will pave the way for a future where AI is deployed responsibly, ethically, and for the benefit of all.

Safeguarding the Rocketfuel of AI: Data

The explosive growth of Artificial Intelligence (AI) hinges on access to vast, diverse, and high-quality datasets. However, current data licensing practices often create significant barriers, hindering innovation and stifling responsible AI development. These barriers include:

- *“Public Release” End User License Agreements (EULA):* These types of open licensing agreements allow companies to purchase proprietary data once and make it available to the general public. These types of EULAs are often-times incompatible with the business model of innovative companies at the forefront of developing proprietary data sets.
- *Uncertainty and Ambiguity*: Vague licensing language creates legal uncertainty, discouraging investment and hindering collaboration.

The Solution:Policies that safeguard and promote companies at the forefront of developing proprietary data sets.

To foster a thriving and responsible AI ecosystem, we need to promote "friendly" data licensing terms that prioritize:

- *Flexibility and Accessibility:* Licenses should offer flexible usage rights, allowing for a wide range of AI applications, including research, development, and commercialization. Fair and reasonable pricing models are also essential.
- *Transparency and Accountability:* Data provenance and usage tracking mechanisms should be implemented to ensure transparency and accountability.
- *Standardization:* Developing industry-wide standards for data licensing can streamline the process and reduce legal uncertainty.

The nature of data licensing carries profound implications for the future of artificial intelligence policy. By establishing business-friendly licensing terms, we can unlock the vast potential of AI, granting wider access to the critical datasets that fuel its development. Equitable participation in the AI landscape is also crucial. Accessible data licensing acts as a leveler, ensuring that smaller companies and research institutions, often constrained by resources, can contribute meaningfully to AI advancements. Beyond innovation and fairness, public trust is paramount. Transparent and ethical data licensing practices are essential in cultivating confidence in AI technologies. Moreover, responsible AI development necessitates clear guidelines and ethical considerations embedded within licensing agreements, ensuring that AI systems are built with integrity. Finally, as AI regulations continue to emerge, business-friendly data licensing will streamline compliance, making it easier for developers to adhere to evolving legal frameworks.

Call to Action: Policymakers should incentivize the development and adoption of business-friendly data licensing terms; we propose the implementation of clear, flexible, and accessible data licensing terms to fuel AI innovation. This includes policies that safeguard and promote companies at the forefront of developing proprietary data sets.

Section 3. Tomorrow.io AI Capabilities Statement

Tomorrow.io delivers an end-to-end weather intelligence platform explicitly designed to handle the complex and evolving requirements of our customers as well as U.S. Government agencies. By combining proprietary satellite data, advanced deep-learning models, and a closed-loop AI framework, Tomorrow.io not only generates real-time, hyper-accurate weather forecasts but also continuously retrains itself to refine predictions based on actual outcomes. This enables the platform to adapt as conditions change, providing actionable weather insights. Here are a sample of use-cases and applications that highlight how Tomorrow.io could transform the mission and operations of U.S. Government Agencies:

Defense & DoD Applications

U.S. defense operations—from large-scale troop deployments to special operations—are inherently weather-dependent. Tomorrow.io's AI-powered space-based platform weaves weather insights into the strategic, operational, and tactical layers of mission execution, providing a decisive information edge in high-risk environments.

- Autonomous Mission Planning
    - Weather-Integrated Strategy: AI forecasts pinpoint dust storms, precipitation, and extreme temperatures, guiding the sequencing and timing of missions to avoid weather-driven failures.
    - Offline Modeling in Field: Units can run disconnected AI simulations that incorporate locally gathered and satellite-derived data, preserving situational awareness even in bandwidth-limited settings.
- Precision Ballistics & Missile Defense
    - Environmental Calibration: Neural networks factor in wind shear, temperature profiles, and air density, heightening accuracy for munitions and missile interception systems powered by the Tomorrow Constellation global atmospheric near-real-time coverage.
- Drone Navigation & Autonomous Convoys
    - Real-Time Rerouting: High-resolution, weather-based route planning decreases the risk of accidents or immobilization caused by flash floods, high winds, or extreme heat.
    - Reduced Operational Delays: Convoys and UAVs remain flexible and on schedule with adaptive weather inputs, minimizing mission vulnerabilities.
- Satellite Link Assurance
    - Uninterrupted Comms: Near real-time precipitation forecasts facilitate automatic channel switching, ensuring continuous command-and-control links during storms.
- Generative AI Mission Briefing
    - Concise, Actionable Reporting: Gale compiles scenario-specific briefs that highlight critical weather alerts, enabling clear directives for commanders and field units.

Weather Advantage: By anticipating environmental disruptions before they escalate, Tomorrow.io's AI-empowered forecasting lowers mission risk, aligns resources efficiently, and bolsters troop protection.

Aviation & Airspace Optimization Applications

Managing U.S. airspace—from civil aviation to military air support—demands accurate and timely weather predictions. Tomorrow.io's platform accommodates both long-term air traffic planning and moment-to-moment operational responses to minimize flight hazards and ensure safety.

- Live Weather Adjustments
  - Dynamic Flight Routing: AI constantly evaluates wind shear, turbulence, icing risks, and traffic congestion, helping agencies like the FAA or DoD avoid preventable delays.
- NAS (National Airspace System) Optimization
  - Predictive Simulation: By integrating historical and real-time datasets (e.g., GDPs, SIDs, STARs), the system forecasts flow constraints, enabling preemptive airspace decongestion.
- Global Performance & Coverage
  - Proprietary Satellite Constellation: A robust pipeline of data improves the training and inference of Tomorrow.io's global models, translating into more precise flight plans and airspace management.
- AI-Enabled Synoptic Monitoring
  - Proactive Detection: Vision-based AI flags severe weather systems, alerting air traffic control centers in time to modify flight paths or issue ground holds.
- Fuel & Emission Efficiency
  - Environmental Compliance: Forecast-optimized routing decreases fuel burn, aligning federal aviation goals with sustainability mandates.
- Generative AI Reporting
  - Clear Communication: Gale condenses complex forecasts into relevant bulletins for aviation officials, facilitating faster coordination among airports, pilots, and command centers.

Weather Advantage: By pairing constant environmental monitoring with adaptive AI forecasting, Tomorrow.io reduces flight disruptions, optimizes airspace utilization, and improves both safety and operational throughput.

Ground Transportation & Traffic Management Applications

For agencies like the Department of Transportation and state highway authorities, weather poses daily challenges in maintaining safe roads, preventing traffic jams, and safeguarding infrastructure. Tomorrow.io's platform offers continuous, AI-driven forecasts that integrate real-time sensor data and historical patterns to anticipate and mitigate road hazards.

- Adaptive Scheduling & Routing
    - Storm-Resilient Operations: Automated planning tools consider temperature drops, flooding potential, and visibility hazards, ensuring minimal disruption to public transit and freight routes.
- Autonomous Safety Enhancements
    - Collision Risk Mitigation: High-fidelity road condition updates feed directly into autonomous vehicle guidance systems, reducing weather-induced accidents.
- Predictive Maintenance
    - Smart Asset Monitoring: Data from IoT sensors embedded in roads, bridges, and tunnels inform tomorrow.io's AI of infrastructure stress points, enabling timely repairs and cost reduction.
- Camera-Vision AI
    - Real-Time Hazard Detection: State DOT cameras enhanced by computer vision detect low visibility, precipitation intensity, and wind events, auto-updating rerouting and public alert systems.

Weather Advantage: Accurate, hyper-local forecasts allow transit agencies and emergency services to keep roads clear, reduce bottlenecks, and allocate maintenance resources more effectively, all by understanding and adapting to imminent weather patterns.

Agriculture & Food Security Applications

Climate volatility threatens U.S. agriculture and overall food security. Tomorrow.io's hyper-local forecasting and AI-driven modeling deliver actionable insights to federal agencies like the USDA, state agricultural commissions, and international food programs.

- Enhanced Parametric Insurance & Risk
  - Better Underwriting: AI-based scenario testing of historical reanalysis conditions improves the reliability of parametric insurance programs, safeguarding farmers' livelihoods.
- AI-Guided Crop Disease Prevention
  - Local & Seasonal Correlations: Neural networks combine weather forecasts with historical outbreak data, facilitating early detection of pests or crop pathogens.
- Autonomous Yield Optimization
  - Data-Driven Best Practices: Tomorrow.io's platform merges sensor inputs and AI simulations to continuously refine growth cycles, maximizing field-level output.
- Precision AI Climate Modeling
  - Targeted Planting & Irrigation: Machine learning pinpoints drought cycles, soil moisture levels, and rainfall trends, ensuring resource allocation is timed and scaled for maximum yield.

Weather Advantage: By predicting and adapting to regional climate variables, Tomorrow.io helps ensure consistent food production, reduces agricultural losses, and fortifies national food security under changing climate conditions.

Wildfire Prediction & Prevention Applications

As wildfires grow increasingly frequent and severe, agencies like the U.S. Forest Service and state emergency management departments rely on early detection and accurate spread modeling. Tomorrow.io's satellite-enabled weather intelligence provides a continuous watch over at-risk areas.

- AI-Enhanced Early Detection
  - Heat Signature Analysis: Neural networks scan satellite thermal data to flag potential ignitions, improving response times and preventing large-scale fires.
- Autonomous Fire Propagation Simulation

    - Dynamic Spread Modeling: Forecasts integrate wind direction, humidity, and terrain data to anticipate wildfire trajectories, guiding evacuation orders and resource deployment.

- AI Smoke Dispersion Forecasting
    - Air Quality Protections: Predictive smoke modeling pinpoints areas where respiratory risks and reduced visibility are imminent, allowing health agencies to issue localized alerts.

Weather Advantage: Real-time meteorological monitoring and data-driven modeling enable faster, more accurate wildfire responses—saving lives, critical infrastructure, and natural resources.

Emergency Management & Disaster Response Applications

From hurricanes to tornadoes, extreme weather events demand coordinated response and advanced warning. Tomorrow.io's AI-driven alerts and proactive planning tools streamline decision-making for agencies such as FEMA and state-level emergency operations centers.

- AI-Driven Hurricane & Tornado Modeling
    - High-Resolution Tracking: Global satellite data powers advanced forecasting to pinpoint landfall, storm intensity, and flood risk, improving warnings and resource staging.
- AI-Powered Outage Prediction
    - Critical Infrastructure Protection: Massive scenario libraries train neural networks to identify power grids at risk, prompting proactive crew deployment.
- Automated First Responder Intelligence
    - Dynamic Coordination: By synthesizing multiple data feeds (radar, crowdsourced reports, sensors), the platform prioritizes rescue operations, reducing chaos during crisis peaks.
- Enhanced AI Tornado Monitoring
    - Early Detection & Nowcasting: Machine learning uses real-time weather anomalies to spot tornadic conditions sooner, giving local communities extra minutes that can save lives.

Weather Advantage: Tomorrow.io's rapid analytics shorten response times, aid in evacuation planning, and reduce the fallout from severe weather events through precise, real-time situational awareness.

Supply Chain & Logistics Optimization

Government agencies must often move critical goods—from medical supplies to heavy equipment—across vast regions, where weather poses a prime source of disruption. Tomorrow.io's predictive analytics and closed-loop AI help orchestrate these operations seamlessly.

- Autonomous Supply Chain Forecasting
  - Preemptive Rerouting: Storm or flood projections trigger automated alerts, prompting shipping routes to shift before bottlenecks develop.
- Historical Analog Comparisons
  - Predictive "What-If": Operators can overlay current weather conditions with historical scenarios to evaluate potential disruptions and quickly enact contingency plans.
- Deep-Learning-Driven Inventory Management
  - Bottleneck Identification: The platform forecasts stock shortages or late deliveries, enabling agencies to reschedule shipments or reallocate resources well ahead of time.
- Multi-Modal AI Logistics Integration
  - Unified Coordination: Road, rail, air, and maritime forecasts converge within Tomorrow.io's neural network, producing holistic scheduling that maximizes throughput and reduces cost.

Weather Advantage: By linking near real-time forecasts and historical event analysis, Tomorrow.io's AI-driven logistics framework promotes timely, flexible operations that withstand weather turbulence—essential for delivering supplies to critical missions.

Conclusion

To conclude, Tomorrow.io stands ready to be a vital partner in advancing the goals of the President's AI Action Plan. Our unique blend of cutting-edge technology, and deep understanding of weather's impact across diverse sectors positions us to make significant contributions. We are confident that our AI-powered weather intelligence platform, coupled with our commitment to innovation and collaboration, will help shape a future where weather-related challenges are mitigated, and AI-driven solutions empower a safer, more resilient, and prosperous society.

We urge the government to consider the recommendations outlined in this document, particularly the focus on empowering small businesses as key drivers of AI innovation, establishing uniform standards for AI model validation and calibration, and promoting business-friendly data licensing practices. By fostering a collaborative environment and embracing forward-thinking policies, we can collectively unlock the transformative potential of AI and ensure its responsible development and deployment for the benefit of this great country.